Data analysis with

jamovi Stats. Open. Now.

Yiu Hong Ko (29-06-2025)


1
Research questions
2
Research design
3
Data collection
4
Data analysis
5
Report

Why Jamovi?

- Free alternative to SPSS
- Open-sourced, open-access
- Flexibility, similar to R (syntax mode)
- Replicability
- Beginner friendly

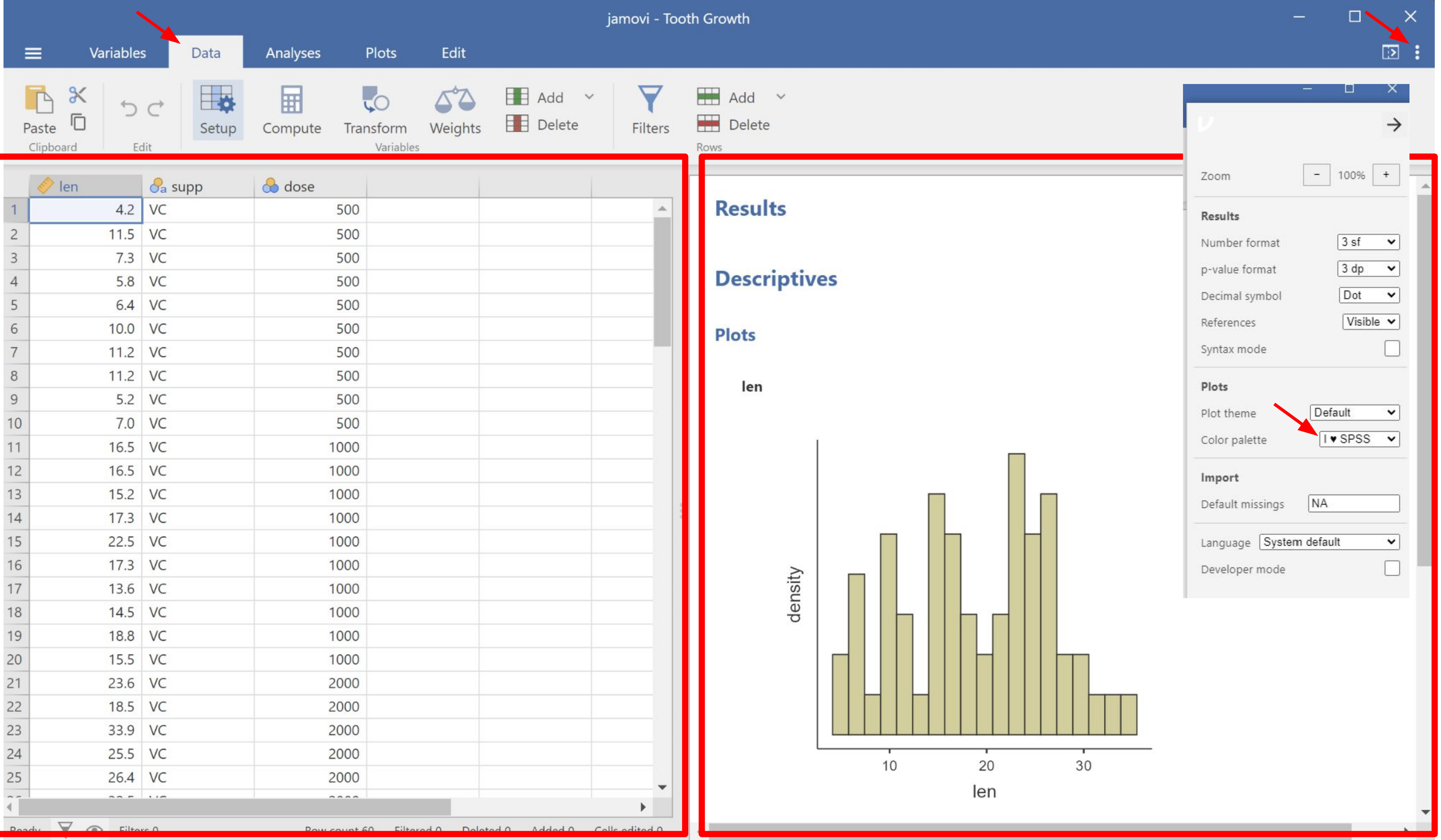
Overview

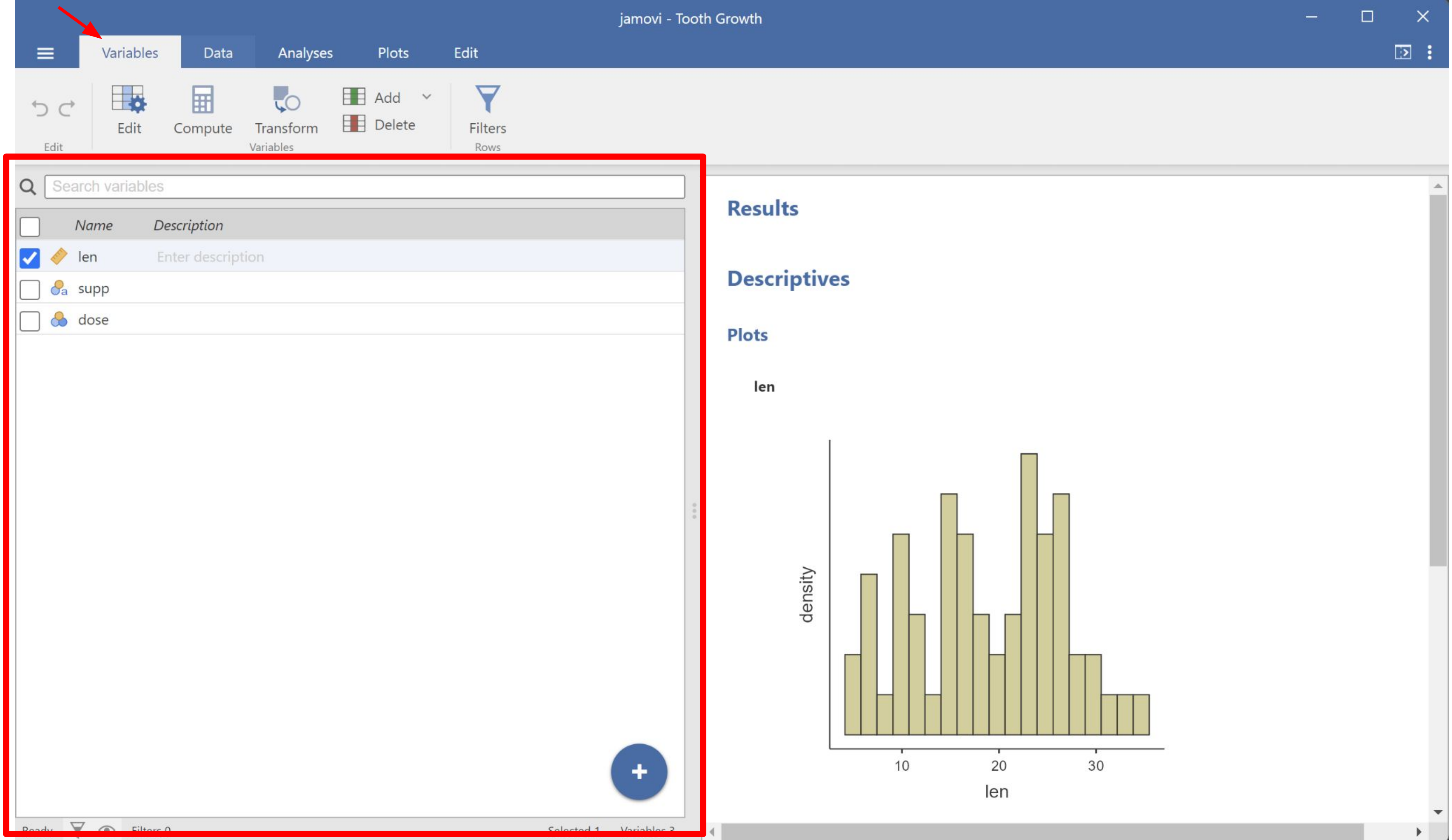
10:00-12:30

1. Load data
2. Data and measure types
3. Compute variables
4. Descriptive statistics
5. Data visualization
6. Filter
7. Transform variables

13:30-16:30

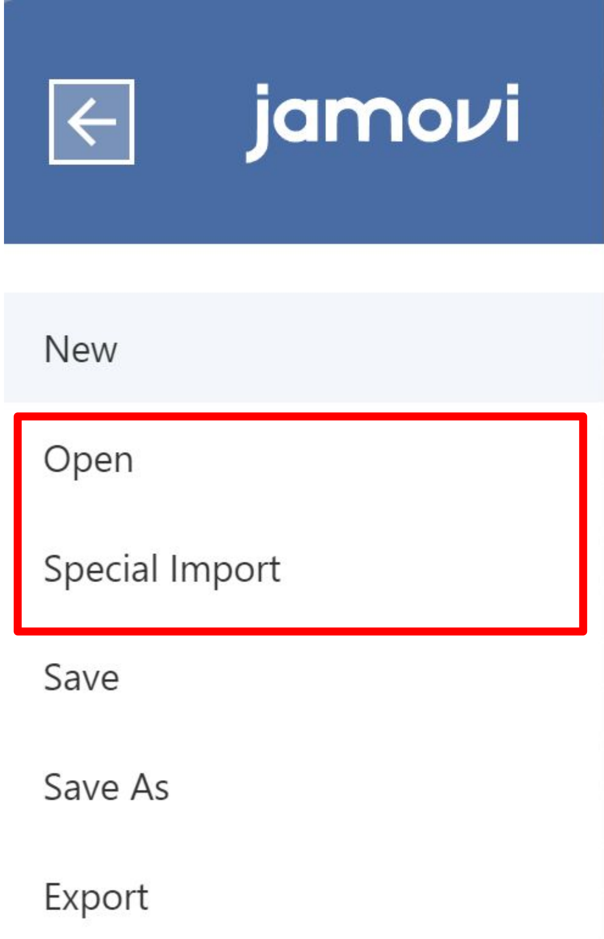
8. Statistics
9. Modules
10. Long format vs. wide format
11. Syntax mode


jamovi - Tooth Growth
Variables
Data
Analyses
Plots
Edit
Paste
Clipboard
Edit
Setup
Compute
Transform
Weights
Add
Delete
Variables
Filters
Add
Delete
Rows
len
supp
dose
1 4.2 VC 500
2 11.5 VC 500
3 7.3 VC 500
4 5.8 VC 500
5 6.4 VC 500
6 10.0 VC 500
7 11.2 VC 500
8 11.2 VC 500
9 5.2 VC 500
10 7.0 VC 500
11 16.5 VC 1000
12 16.5 VC 1000
13 15.2 VC 1000
14 17.3 VC 1000
15 22.5 VC 1000
16 17.3 VC 1000
17 13.6 VC 1000
18 14.5 VC 1000
19 18.8 VC 1000
20 15.5 VC 1000
21 23.6 VC 2000
22 18.5 VC 2000
23 33.9 VC 2000
24 25.5 VC 2000
25 26.4 VC 2000
Results
Descriptives
Plots
len
density
10
20
30
len
Zoom
100%
Results
Number format
3 sf
p-value format
3 dp
Decimal symbol
Dot
References
Visible
Syntax mode
Plots
Plot theme
Default
Color palette
I ♥ SPSS
Import
Default missings
NA
Language
System default
Developer mode


jamovi - Tooth Growth
Variables
Data
Analyses
Plots
Edit
Edit
Compute
Transform
Add
Delete
Filters
Edit
Variables
Rows
Search variables
Name
Description
len
Enter description
supp
dose
Results
Descriptives
Plots
len
density
10
20
30
len

1. Load data

- Open: Loading one single file
    - Data Library
- Special Import: Loading and stacking multiple files
    - for combining datasets (e.g., one dataset for each participant)

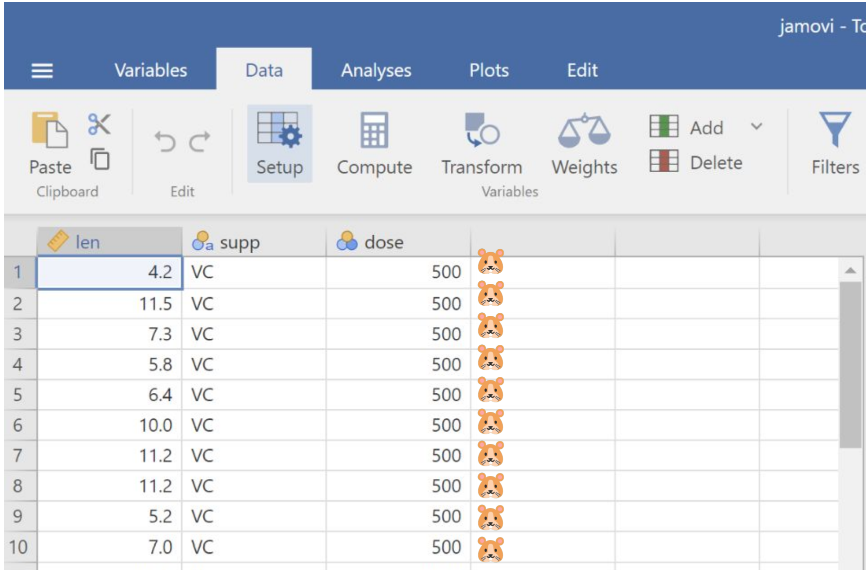


[Guinea pig - Wikipedia](https://en.wikipedia.org/wiki/Guinea_pig)

Exercise 1: load the Tooth Growth dataset

The `Tooth Growth` example contains the length of teeth from guinea pigs (the `len` column) fed different dosages (the `dose` column) of supplements: vitamin C or orange juice (recorded in the `supp` column as `VC` and `OJ`).

	500	1000	2000
VC💊	🐹🐹🐹🐹🐹 🐹🐹🐹🐹🐹	🐹🐹🐹🐹🐹 🐹🐹🐹🐹🐹	🐹🐹🐹🐹🐹 🐹🐹🐹🐹🐹
OJ🍊	🐹🐹🐹🐹🐹 🐹🐹🐹🐹🐹	🐹🐹🐹🐹🐹 🐹🐹🐹🐹🐹	🐹🐹🐹🐹🐹 🐹🐹🐹🐹🐹



2. Data type and measure types

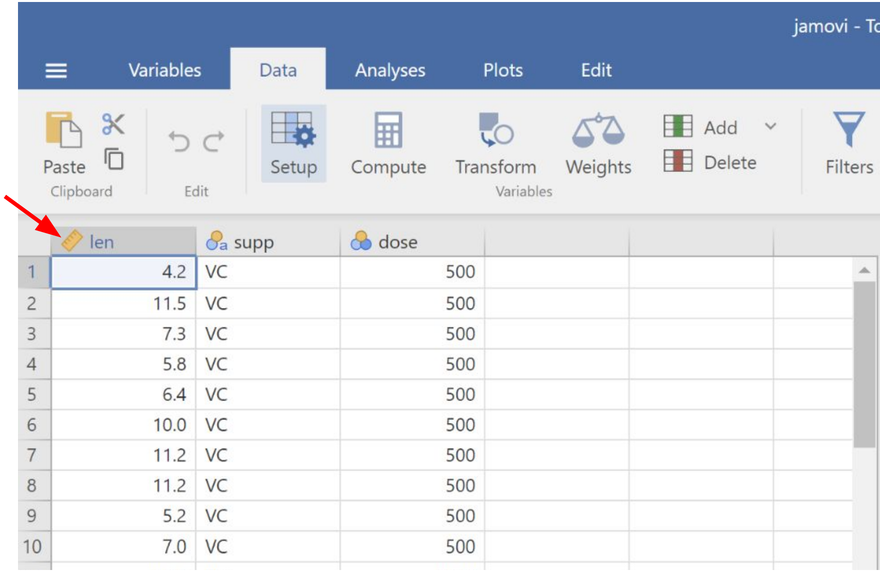
Data variables can be one three data types:

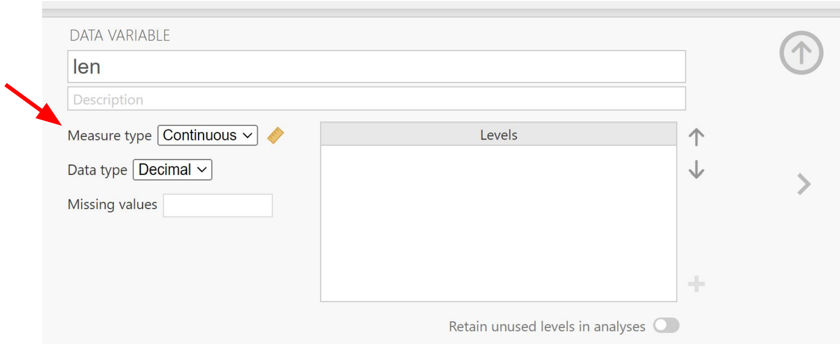
```
Integer
Decimal
Text
```

and one of four measure types:

```
Nominal
Ordinal
Continuous
ID
```

They are editable (to a certain extent)





3. Compute variables

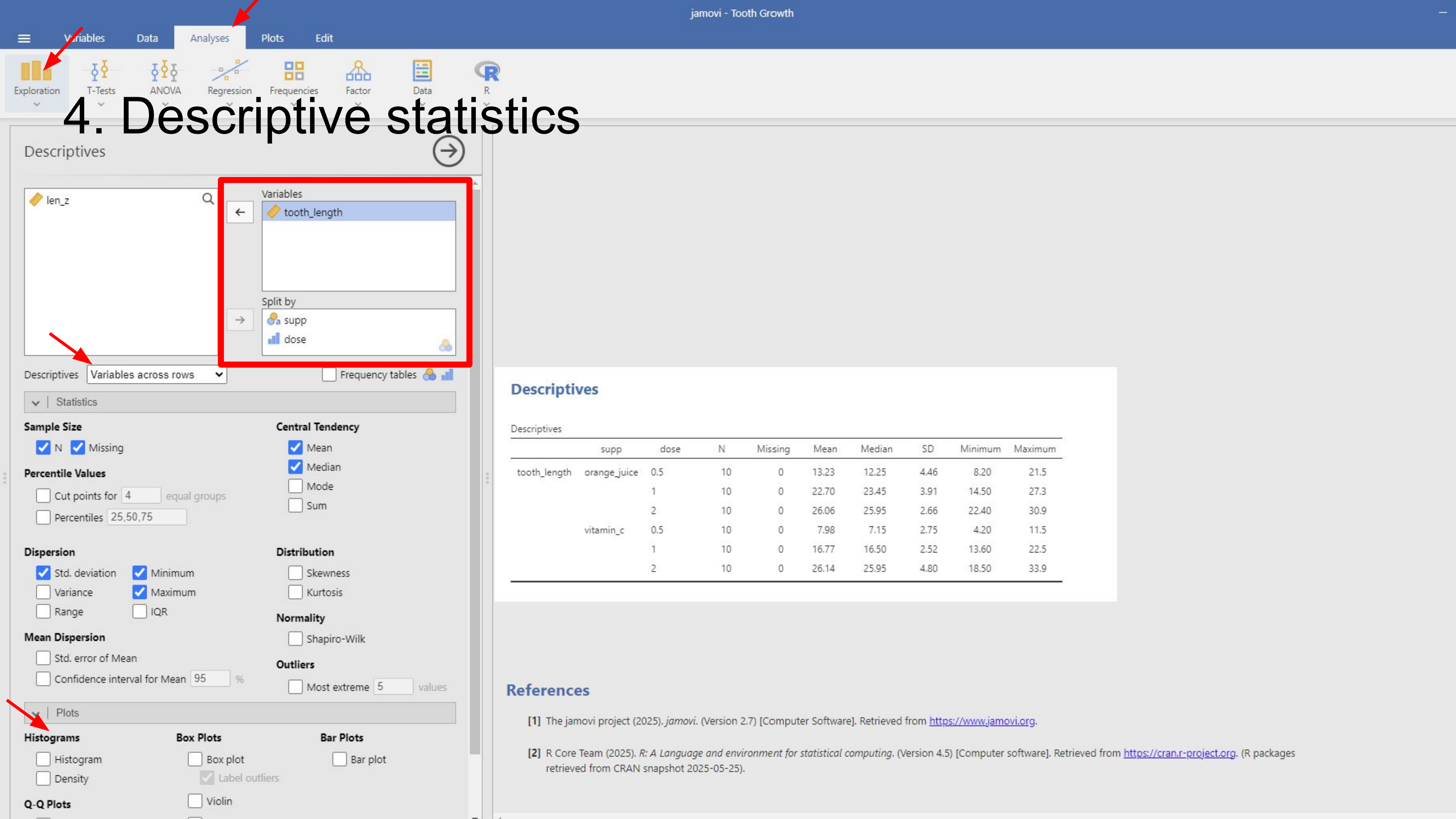
E.g.,

- Arithmetic:
    - A + B
    - CRT-SRT
- Functions (across columns):
    - MEAN(A, B)
    - LOG10(len)
- Variable function (across rows):
    - VMEAN(len, group_by=dose)
- Statistics, e.g., z-scores
    - (dose - VMEAN(dose)) / VSTDEV(dose)
    - Z(dose)

Exercise 2: Label and compute variables

1. Rename *len* to *tooth_length*
2. Label *OJ* as *orange_juice*, and *VC* as *vitamin_c*
3. Change the variable type of *dose* into ordinal
4. Reorder the levels of *dose* as 2000, 1000, 500 (instead of 500, 1000, 2000)
5. Label the levels of *dose* in mg (Hint: 1mg = 1000 mcg)
6. Compute *len_z* the z-score of len (Hint: *Z(tooth_length)*)

4. Descriptive statistics



Descriptives

Descriptives

	supp	dose	N	Missing	Mean	Median	SD	Minimum	Maximum
tooth_length	orange_juice	0.5	10	0	13.23	12.25	4.46	8.20	21.5
		1	10	0	22.70	23.45	3.91	14.50	27.3
		2	10	0	26.06	25.95	2.66	22.40	30.9
	vitamin_c	0.5	10	0	7.98	7.15	2.75	4.20	11.5
		1	10	0	16.77	16.50	2.52	13.60	22.5
		2	10	0	26.14	25.95	4.80	18.50	33.9

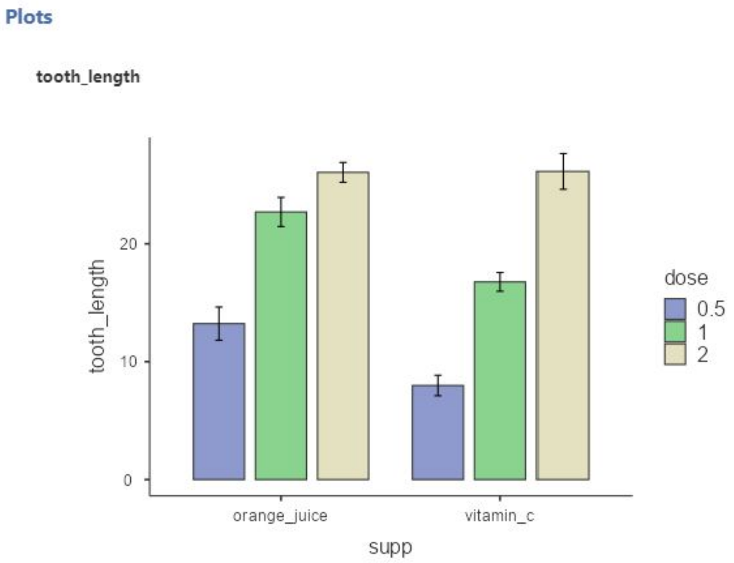
References

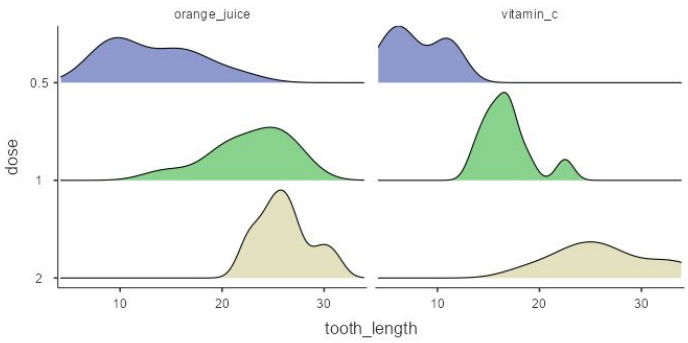
[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org.

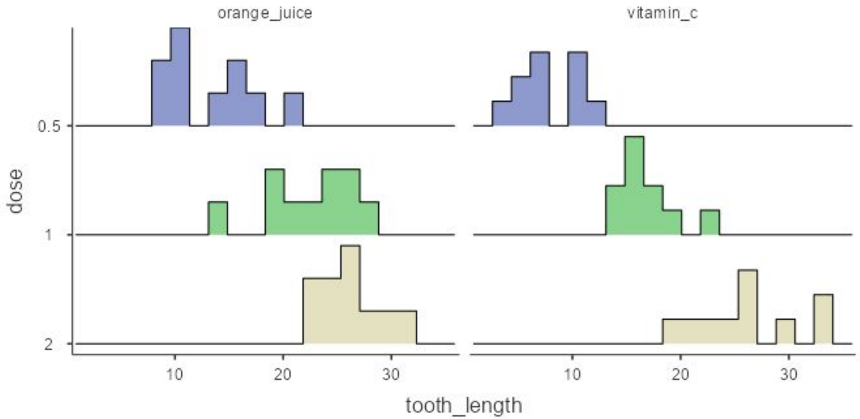
[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25).

5. Data visualization

- bar charts
- histograms
- density plots

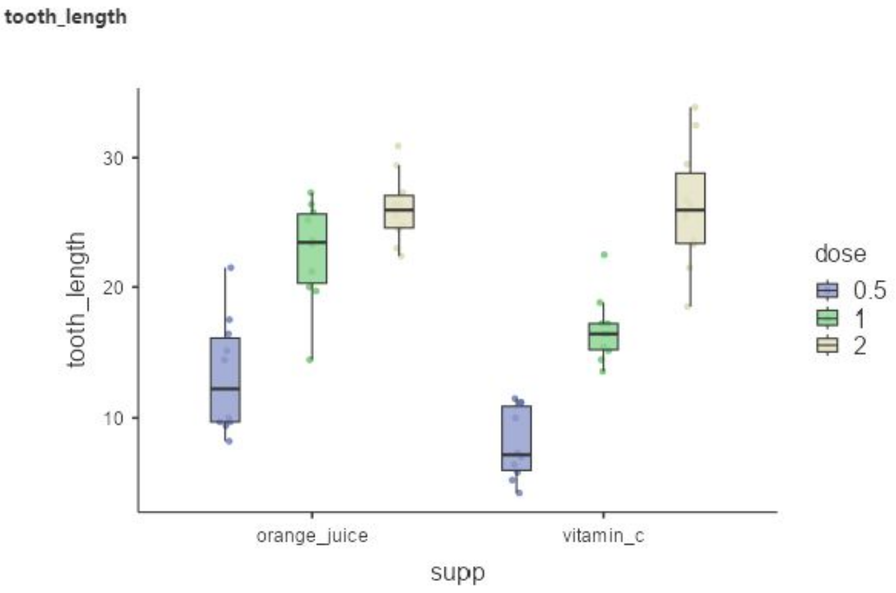


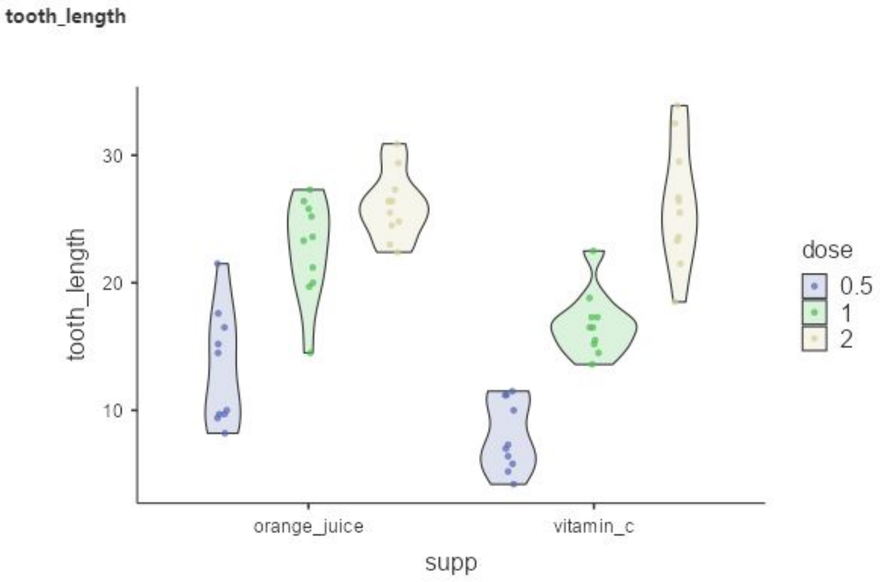




5. Data visualization

- box plots
- violin plots



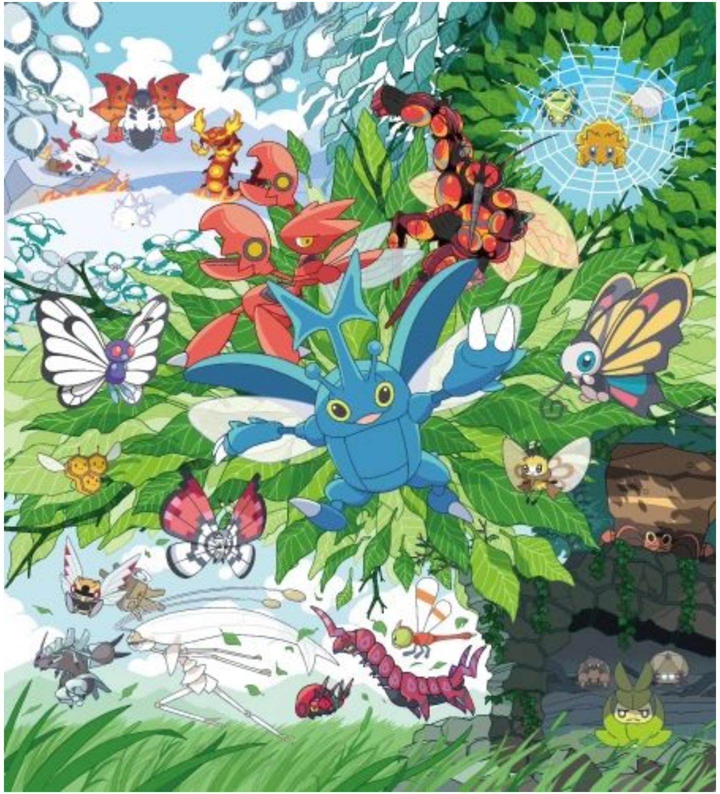


Exercise 3: Bugs!

The extent to which men and women want to kill arthropods that vary in freighteningness (low, high) and disgustingness (low, high).
Each participant rates their attitudes towards all anthropods.

With 372 rows and 6 variables

- subject. Subject ID
- gender. Female or Male
- region. Region of the world the participant was from.
- education. Level of education.
- desire. The desire to kill an arthropod indicated on a scale from 0 to 10, In the following conditions:
  - **LDLF**: low freighteningness and low disgustingness;
  - **LFHD**: low freighteningness and high disgustingness;
  - **HFHD**: high freighteningness and low disgustingness;
  - **HFHD**: high freighteningness and high disgustingness



Bug (Pokémon type) | Nintendo | Fandom

https://www.indrapatil.com/ggstatsplot/reference/bugs_long.html

Ryan, R. S., Wilde, M., & Crist, S. (2013). Compared to a small, supervised lab experiment, a large, unsupervised web-based experiment on a previously unknown effect has benefits that outweigh its potential costs. *Computers in Human Behavior*, *29*(4), 1295-1301.

Exercise 3: Bugs!

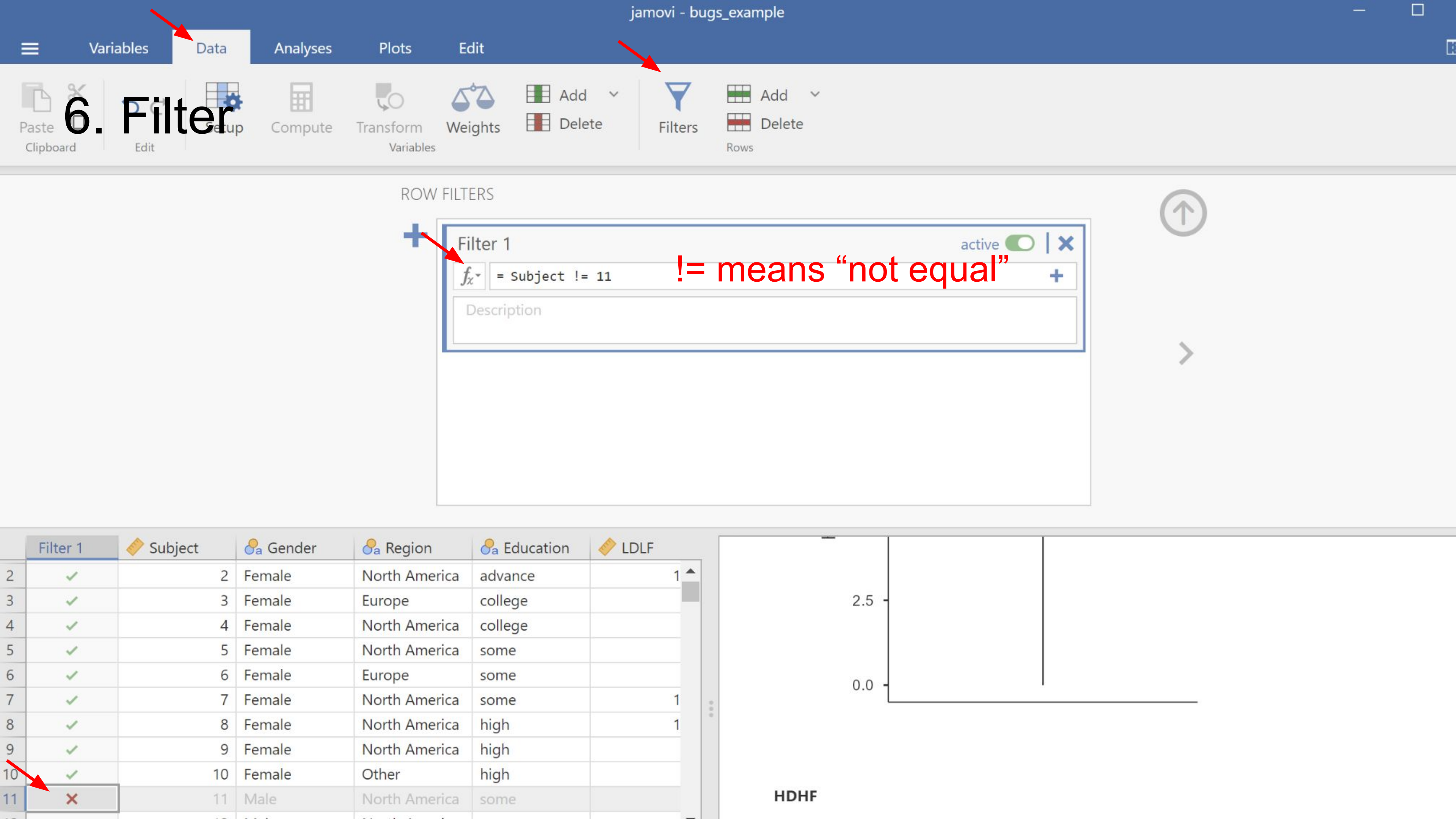
1. Load the Bugs dataset from the data library
2. Check the data and measure types; correct them if needed
3. Compute a *desire_to_kill* rating by averaging the ratings across all conditions
4. Explore the summary statistics of this *desire_to_kill* variable
5. Based on data visualization, determine if *desire_to_kill* differs across:
    a. Gender
    b. Education
    c. Region

6. Categorize individuals (rows) into high vs. low desire_to_kill

6. Filter

- With a clear rationale, outliers / invalid data could be excluded from the analysis
- E.g., in the Bugs dataset, subject 11 has very low ratings across all conditions
    - 0, 2.5, 0, 0
- It could be excluded with the filter function
- As with other changes in the data, outputs will be automatically updated

6. Filter

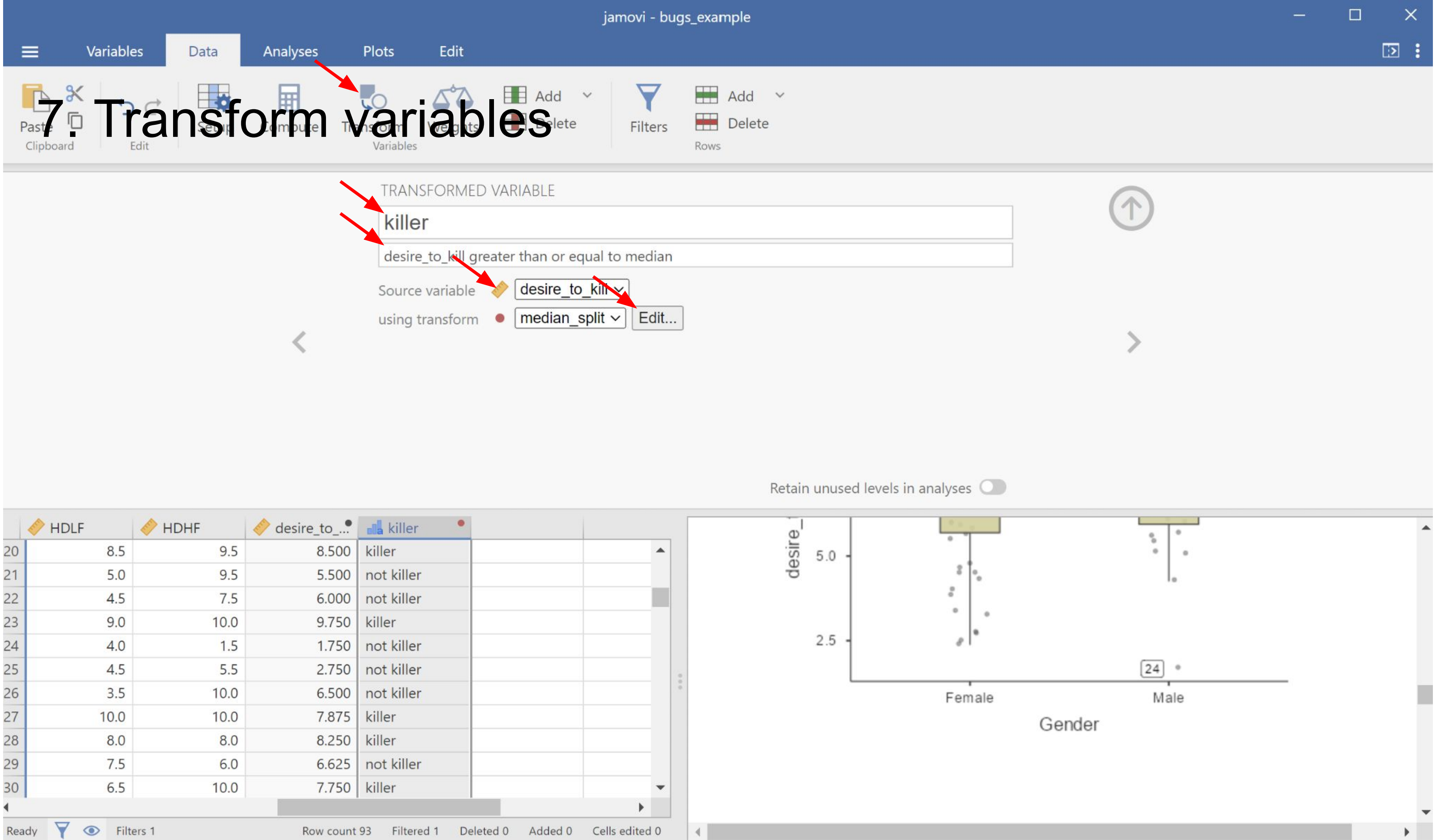


!= means “not equal”

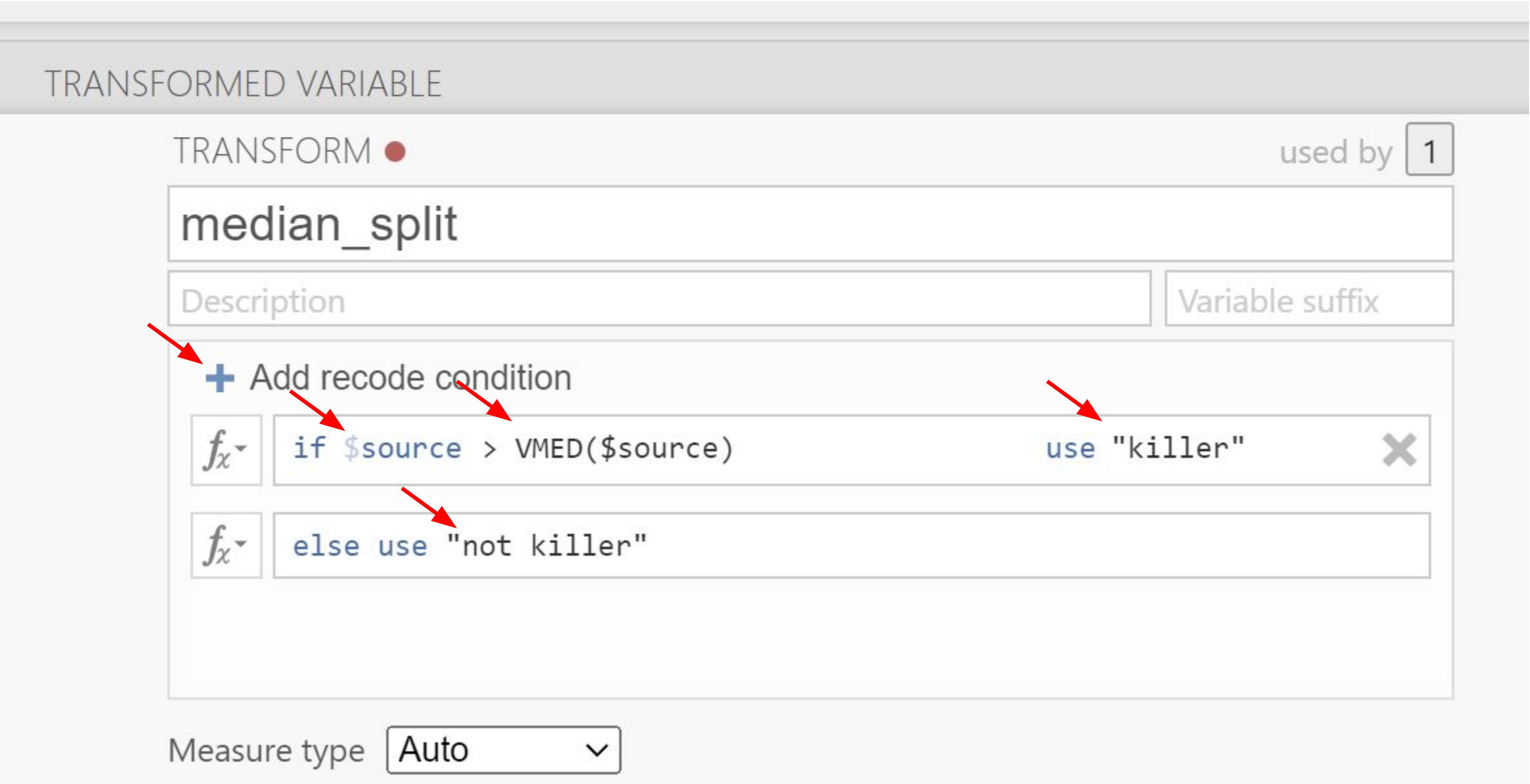
7. Transform variables

- Unlike *compute*, which could work on multiple variables, *transform* works on one variable. However, it could reuse transformation functions.
- Typically used for recoding variables, or with functions applied to multiple variables repeatedly
- E.g., median split could be applied to dichotomize variables (usually not recommended though)
    - In the BUGS dataset, an individual could be categorized as “*killer’* if the *desire_to_kill* rating is larger or equal to the median, and *“not killer”* if not

7. Transform variables



7. Transform variables



8. Statistics!

- Unlike SPSS, Javmovi does not have as many analysis options
- However, the outputs of the analysis are often clearer and follow the APA format
- By design, the ways to execute them are similar to SPSS

8. Statistics! - Independent t test

Does *desire_to_kill* differ across genders?

Independent Samples T-Test

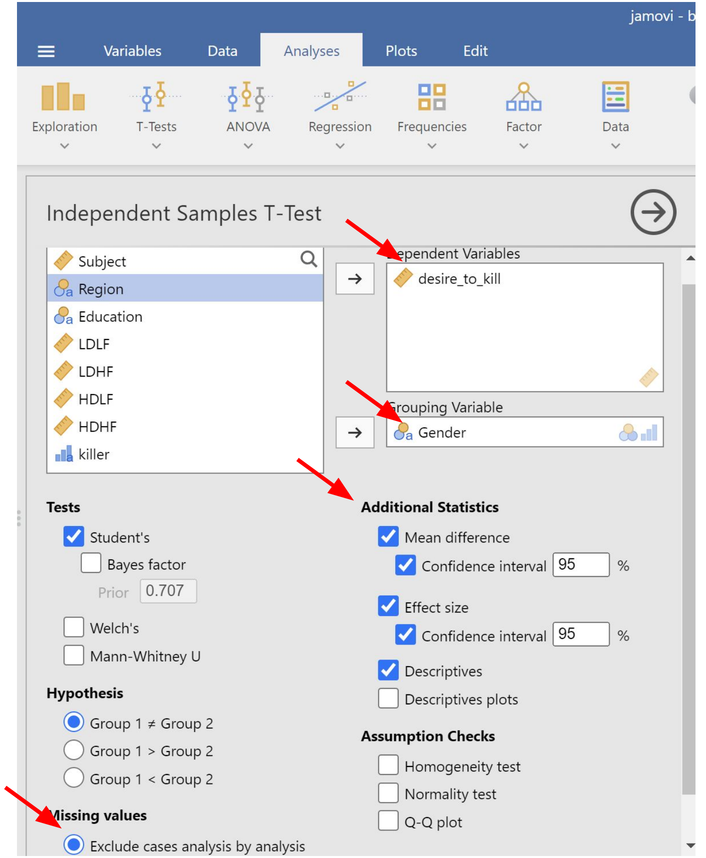
Independent Samples T-Test

		Statistic	df	p	Mean difference	SE difference	95% Confidence Interval Lower	95% Confidence Interval Upper		Effect Size	95% Confidence Interval Lower	95% Confidence Interval Upper
desire_to_kill	Student's t	0.998	85.0	.321	0.487	0.488	−0.483	1.46	Cohen's d	0.227	−0.221	0.673

Note. H_a $\mu_{\text{Female}} \neq \mu_{\text{Male}}$

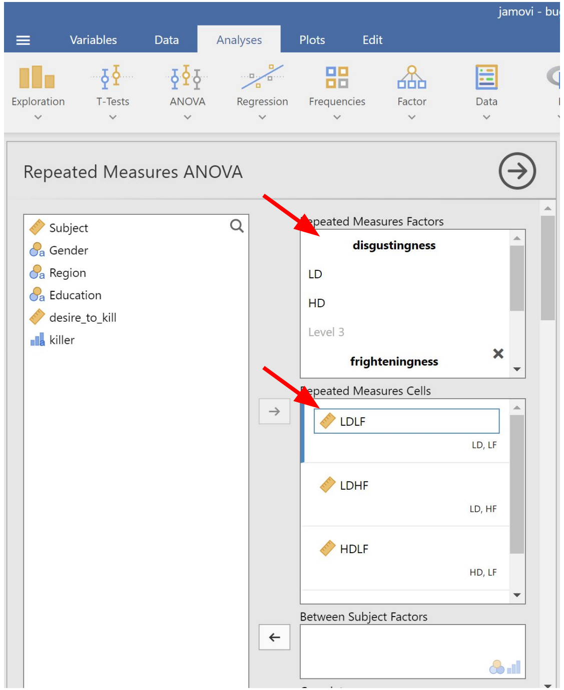
Group Descriptives

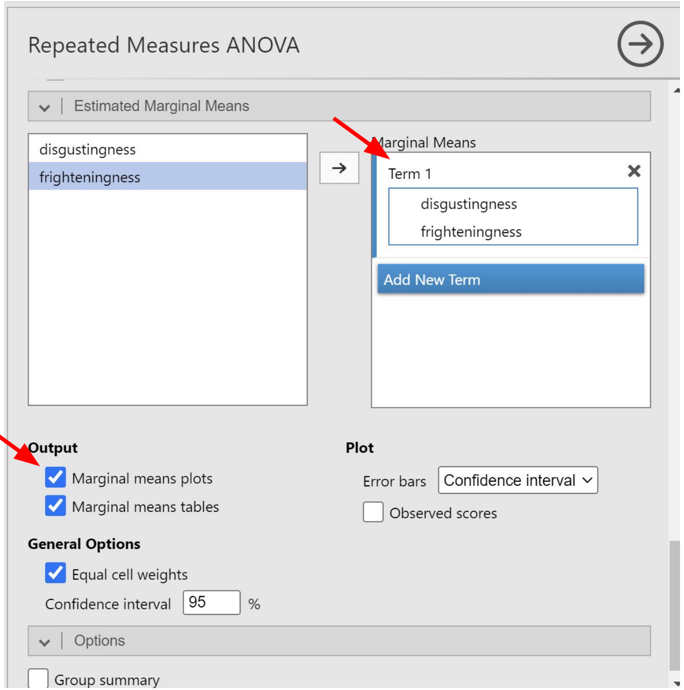
	Group	N	Mean	Median	SD	SE
desire_to_kill	Female	58	7.02	7.56	2.24	0.294
	Male	29	6.53	6.88	1.95	0.362



8. Statistics! - Repeated measures ANOVA

Do *disgustingness* and *frighteningness* affect *desire_to_kill*?





8. Statistics! - Repeated measures ANOVA

Repeated Measures ANOVA

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
disgustingness	48.75	1	48.75	12.18	< .001
Residual	348.37	87	4.00		
frighteningness	177.56	1	177.56	41.63	< .001
Residual	371.07	87	4.27		
disgustingness ✻ frighteningness	6.55	1	6.55	2.15	.146
Residual	264.58	87	3.04		

Note. Type 3 Sums of Squares

[3]

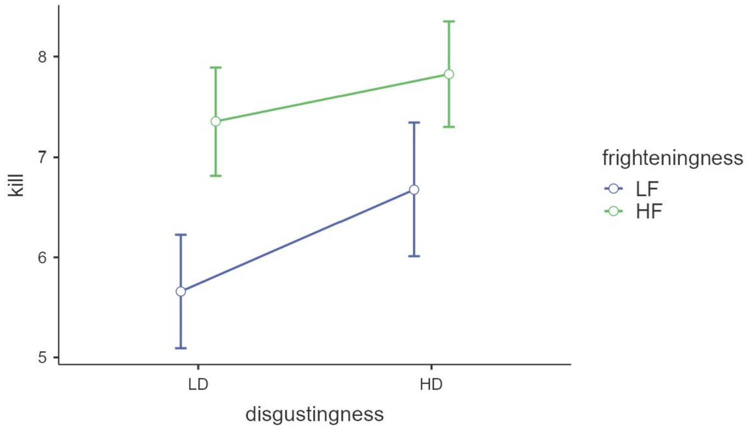
Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Residual	1595	87	18.3		

Note. Type 3 Sums of Squares

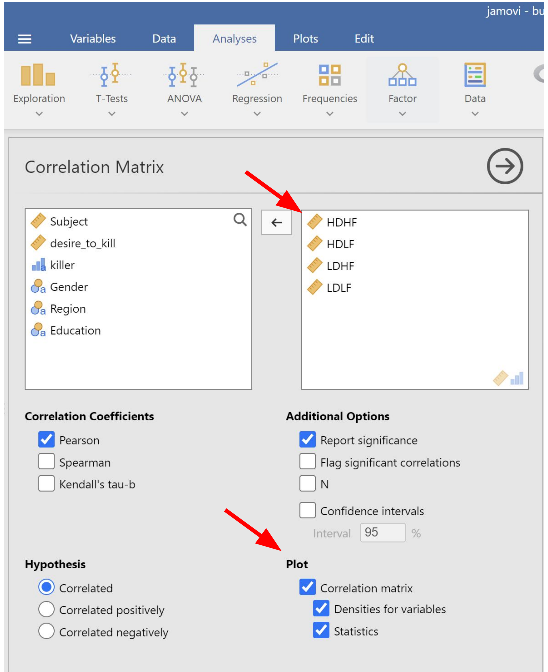
Estimated Marginal Means

disgustingness ✻ frighteningness



8. Statistics! - Correlation

Are the ratings across four conditions correlated?

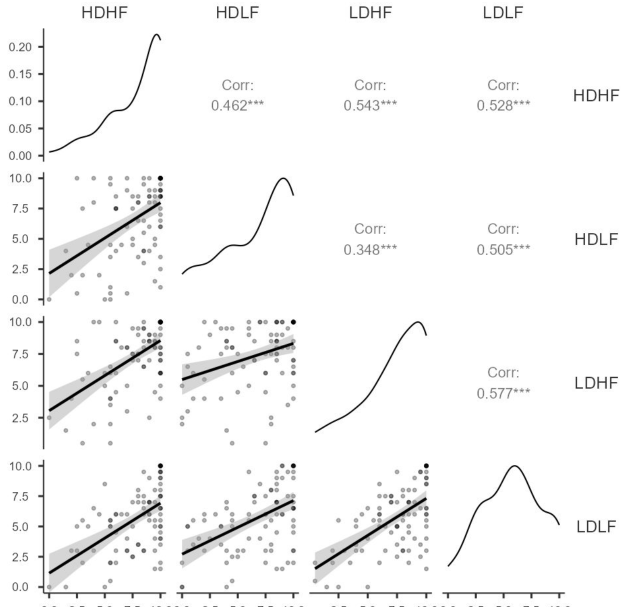


Correlation Matrix

Correlation Matrix

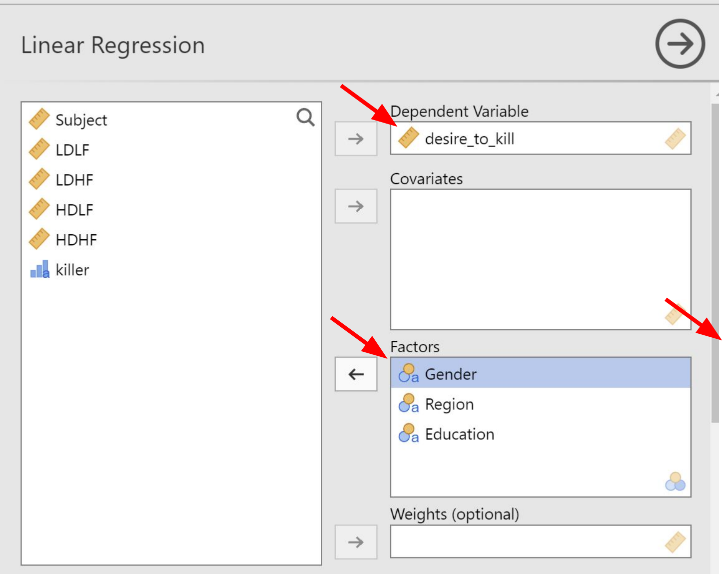
		HDHF	HDLF	LDHF	LDLF
HDHF	Pearson's r	—			
	df	—			
	p-value	—			
HDLF	Pearson's r	0.462	—		
	df	88	—		
	p-value	< .001	—		
LDHF	Pearson's r	0.543	0.348	—	
	df	88	87	—	
	p-value	< .001	< .001	—	
LDLF	Pearson's r	0.528	0.505	0.577	—
	df	90	89	89	—
	p-value	< .001	< .001	< .001	—

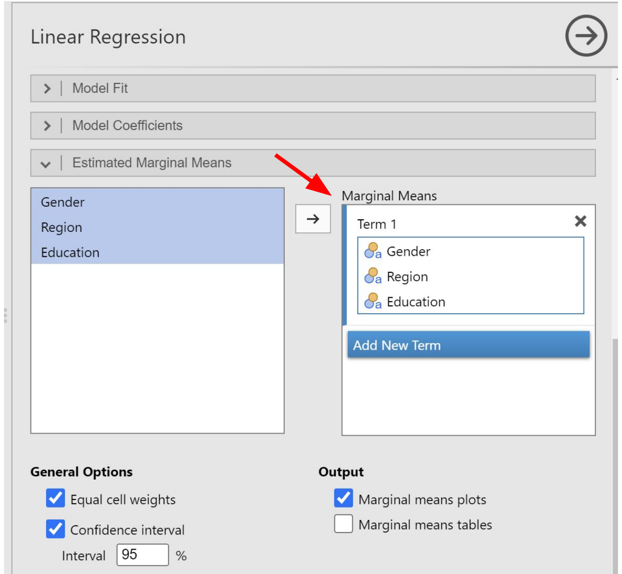
Plot



8. Statistics! - Regression

Do gender, region, and education predict overall *desire_to_kill*?





Linear Regression

Model Fit Measures

Model	R	R^2
1	0.399	0.159

Note. Models estimated using sample size of N=86

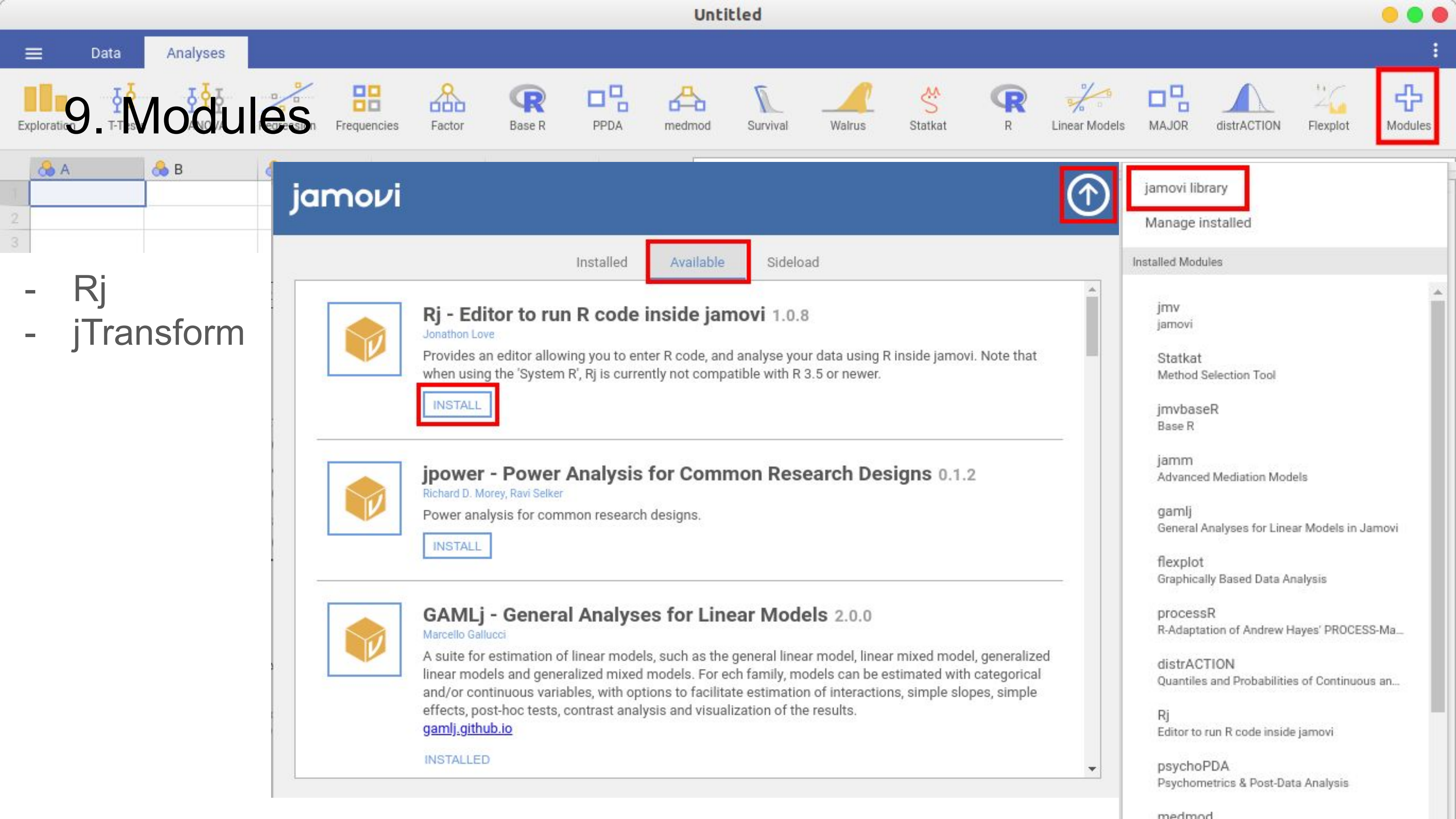
Model Coefficients - desire_to_kill

Predictor	Estimate	SE	t	p
Intercept [a]	3.093	2.982	1.037	.303
Gender:				
Male – Female	−0.500	0.490	−1.020	.311
Region:				
Europe – Australia	−0.967	2.245	−0.431	.668
North America – Australia	0.782	2.125	0.368	.714
Other – Australia	1.833	2.415	0.759	.450
South America – Australia	3.875	2.958	1.310	.194
Education:				
college – advance	3.800	2.322	1.636	.106
high – advance	4.000	2.183	1.832	.071
less – advance	3.518	2.149	1.637	.106
partial – advance	2.250	2.573	0.874	.385
some – advance	2.907	2.125	1.368	.175

[a] Represents reference level

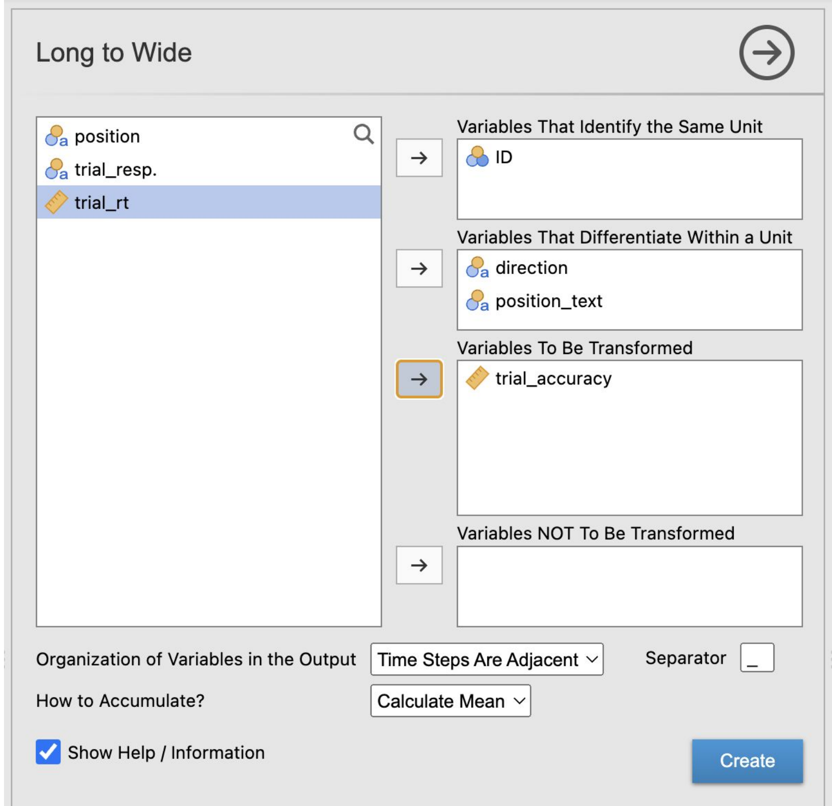
9. Modules

- Rj
- jTransform



10. Long format and wide format

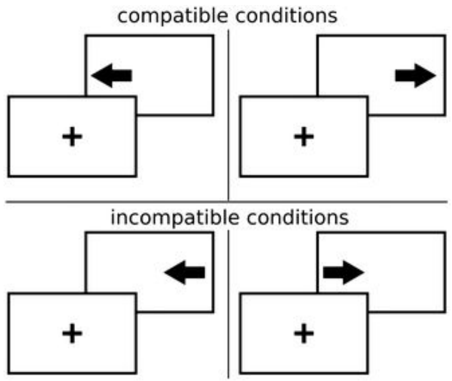
- Data are often stored in the long or wide format
- E.g.,
  - behavioral tasks with multiple trials store data in the long format
  - statistical packages sometimes expect data in the wide format
- Jamovi doesn’t have a function for that but the module *jTransfrom* can do that



Exercise 4: Simon task

Download the data sets *simon_data* and *gonogo_data* from this link https://drive.google.com/drive/folders/12meW_MLh03t6F_aCJFA02S9Xd0tXZd_B?usp=drive_link

Open *simon_data* and *Special Import* all files



Stoet, G. (2017). Sex differences in the Simon task help to interpret sex differences in selective attention. *Psychological research*, *81*(3), 571-581.

Exercise 4: Simon task

1. Convert positions coordinates to “left” or “right” (HINT: [-0.25, 0] = “left”)
2. Compute *trial_accuracy* (HINT: *VALUE((direction == trial_resp.))*)
3. Convert the long format data file into wide format, for accuracy and conditions respectively, and merge them into one

Accuracy by condition (direcion × position)

	ID	direction	position	trial_resp.	trial_rt	trial_acc...	position_...
1	1	right	[0.25, 0]	right	0.369	1	right
2	1	right	[0.25, 0]	right	0.391	1	right
3	1	left	[0.25, 0]	left	0.468	1	right
4	1	right	[0.25, 0]	left	0.393	0	right
5	1	right	[0.25, 0]	right	0.351	1	right
6	1	left	[-0.25, 0]	right	0.390	0	left
7	1	left	[0.25, 0]	right	0.456	0	right
8	1	right	[0.25, 0]	right	0.497	1	right
9	1	right	[-0.25, 0]	right	0.486	1	left
10	1	right	[-0.25, 0]	right	0.504	1	left
11	1	right	[0.25, 0]	right	0.449	1	right
12	1	left	[0.25, 0]	left	0.410	1	right

Long format:
Each measure is a row

	ID	trial_acc...	trial_acc...	trial_acc...	trial_acc...
1	1	0.92	0.88	0.92	0.96
2	2	0.84	0.88	0.84	0.92
3	3	0.84	0.92	0.88	0.88
4	4	0.88	0.96	0.92	0.92
5	5	0.96	0.92	0.92	0.92
6	6	0.92	0.96	0.84	0.88
7	7	0.96	0.88	0.84	0.88
8	8	0.92	0.92	0.84	0.92
9	9	0.88	0.88	0.96	0.92
10	10	0.88	0.92	0.92	0.84

Wide format:
Each participant is a row

RT by condition (direcion × position × trial_accuracy)

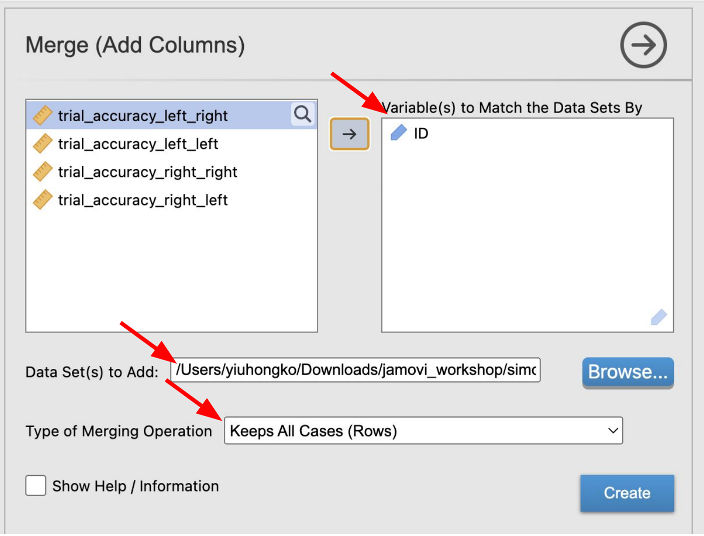
	ID	direction	position	trial_resp.	trial_rt	trial_acc...	position_...
1	1	right	[0.25, 0]	right	0.369	1	right
2	1	right	[0.25, 0]	right	0.391	1	right
3	1	left	[0.25, 0]	left	0.468	1	right
4	1	right	[0.25, 0]	left	0.393	0	right
5	1	right	[0.25, 0]	right	0.351	1	right
6	1	left	[-0.25, 0]	right	0.390	0	left
7	1	left	[0.25, 0]	right	0.456	0	right
8	1	right	[0.25, 0]	right	0.497	1	right
9	1	right	[-0.25, 0]	right	0.486	1	left
10	1	right	[-0.25, 0]	right	0.504	1	left
11	1	right	[0.25, 0]	right	0.449	1	right
12	1	left	[0.25, 0]	left	0.410	1	right

Long format:
Each measure is a row

Wide format:
Each participant is a row

	ID	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...
1	1	0.513	0.504	0.390	0.405	0.440	0.442	0.495	0.476
2	2	0.555	0.520	0.314	0.345	0.373	0.376	0.411	0.430
3	3	0.463	0.455	0.442	0.384	0.436	0.386	0.471	0.478
4	4	0.463	0.497	0.384	0.438	0.416	0.426	0.541	0.483
5	5	0.392	0.461	0.401	0.398	0.365	0.433	0.517	0.512
6	6	0.476	0.507	0.464	0.461	0.377	0.460	0.438	0.501
7	7	0.422	0.473	0.347	0.381	0.463	0.417	0.464	0.503
8	8	0.385	0.497	0.345	0.440	0.437	0.424	0.441	0.487
9	9	0.521	0.446	0.395	0.369	0.295	0.388	0.397	0.429
10	10	0.507	0.507	0.396	0.406	0.438	0.393	0.479	0.508

Merge



	ID	trial_acc...	trial_acc...	trial_acc...	trial_acc...	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_l...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...	trial_rt_ri...
1	1	0.92	0.88	0.92	0.96	0.513	0.504	0.390	0.405	0.440	0.442	0.495	0.476
2	10	0.88	0.92	0.92	0.84	0.507	0.507	0.396	0.406	0.438	0.393	0.479	0.508
3	2	0.84	0.88	0.84	0.92	0.555	0.520	0.314	0.345	0.373	0.376	0.411	0.430
4	3	0.84	0.92	0.88	0.88	0.463	0.455	0.442	0.384	0.436	0.386	0.471	0.478
5	4	0.88	0.96	0.92	0.92	0.463	0.497	0.384	0.438	0.416	0.426	0.541	0.483
6	5	0.96	0.92	0.92	0.92	0.392	0.461	0.401	0.398	0.365	0.433	0.517	0.512
7	6	0.92	0.96	0.84	0.88	0.476	0.507	0.464	0.461	0.377	0.460	0.438	0.501
8	7	0.96	0.88	0.84	0.88	0.422	0.473	0.347	0.381	0.463	0.417	0.464	0.503
9	8	0.92	0.92	0.84	0.92	0.385	0.497	0.345	0.440	0.437	0.424	0.441	0.487
10	9	0.88	0.88	0.96	0.92	0.521	0.446	0.395	0.369	0.295	0.388	0.397	0.429

Exercise 4: Does congruence affect accuracy and RT?

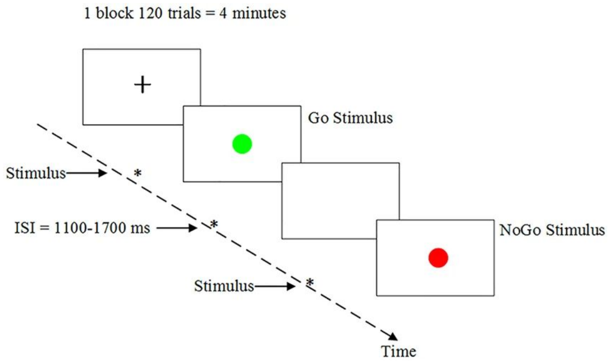
1. Compute a variable that indicates congruence
2. Create a wide format of the data
3. Use a statistical test to determine if there’s any difference between congruence conditions in accuracy or RT

Final challenge!

Ten students completed a go/no-go task. To analyze their data, you need to:

1. Preprocess data
   - Import data
   - Rename variables and values if necessary
   - Compute trial_accuracy
   - Compute trial_outcome (Hit, FA, Miss, CR)

2. Data analysis
   - Does accuracy differ across go and no-go condition?
   - Does RT differ across hits and false alarm?
   - Find out how to calculate d’ and criterion c
   - Are the values of d’ and c significantly different from 0?



Guo, Z., Chen, R., Liu, X., Zhao, G., Zheng, Y., Gong, M., & Zhang, J. (2018). The impairing effects of mental fatigue on response inhibition: An ERP study. *PloS one*, *13*(6), e0198206.

How to compute *trial_outcome*

COMPUTED VARIABLE

trial_outcome

Description

Formula

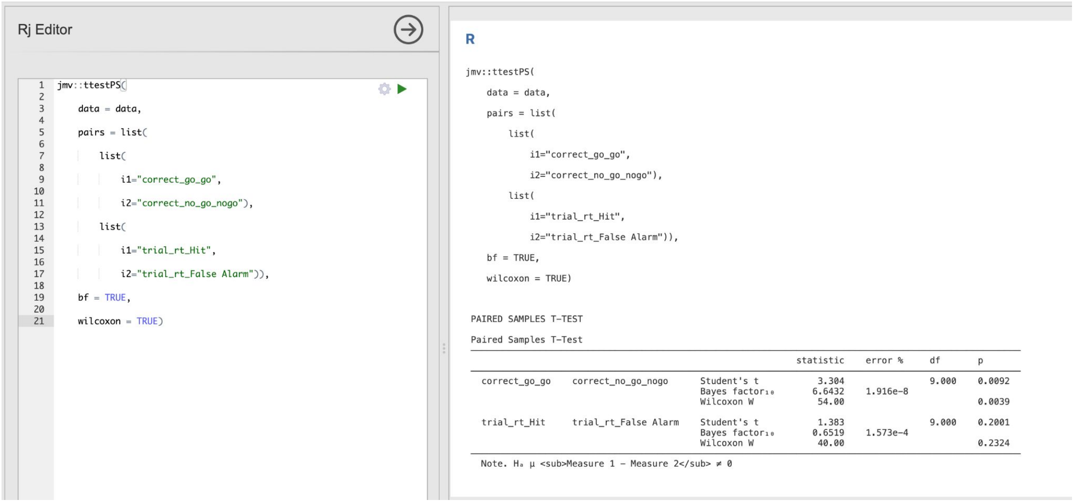
```
= IF(trial_type == 'go' and trial_resp. == 'press',
'Hit', IF(trial_type == 'go' and trial_resp. == 'none',
'Miss', IF(trial_type == 'nogo' and trial_resp. ==
'none', 'Correct Rejection', 'False Alarm')))
```

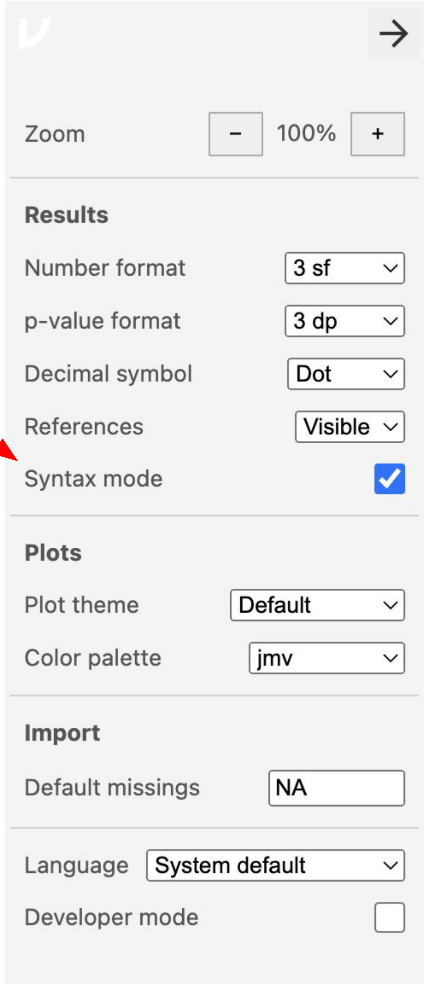
Retain unused levels in analyses

11. Jamovi syntax in R

Procedures on Jamovi could be recreated using the underlying R codes, which is available if Syntax mode enabled.

Find some code above the outputs in the result panel, and copy that to the Rj editor and run it!





If you’re interested in R

R: CRAN

RStudio: RStudio IDE | The Premier Free IDE for R

RStudio is an Integrated Development Environment (IDE) for R

Resources

https://youtu.be/mZomeS0tLxY?si=YBIlvCYcFlOqeWh3

https://lsj.readthedocs.io/en/latest/

community resources - jamovi

Statistics in jamovi – Simple Book Publishing

jamovi Documentation